

Convergence Pressure

Measuring AI-Mediated Cultural Homogenization in Iterated Creation

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Abstract. Generative AI raises the creativity of an individual while lowering the diversity of the crowd (Doshi & Hauser, 2024); models retrained on their own output collapse (Shumailov et al., 2024). We join the two into one dynamical question: when a shared model mediates an **iterated** creative process, does a population’s semantic diversity decay over generations, and what drives it? We run a controlled, reproducible experiment in which a fixed pool of diverse creator personas produces one artifact per generation under four conditions: writing alone, writing with a static AI advisor, writing with an advisor that reflects the population’s own recent output back at it, and the same reflective loop with a panel of diverse advisors. The headline result is a **dissociation**: AI assistance *per se* leaves population diversity flat (100–102% retained over six generations), but the **reflective loop**, the AI echoing the crowd’s recent hits, drives an anisotropy-controlled decline of about 10–12%. The obvious fix fails: a panel of **diverse** AI advisors, which preserves variety in a single round, does **not** prevent the collapse under iteration (it loses slightly more). We report the effect under length-matching and anisotropy controls, keep the metric-dependence and this negative result in view, and close with a philosophical account of why a contracting measure of variety is evidence about, not proof of, a cultural monoculture.

1 Introduction

The motivating tension is now well documented at the level of a **single** interaction. Doshi and Hauser [1] gave writers access to generative-AI story ideas and found a scissors: individual stories were rated more creative and better written, yet the set of stories became **more similar to one another**. Padmakumar and He [2] report the same compression of content diversity when people write with language models. Separately, in the pure-retraining setting, Shumailov et al. [3] show that a model trained recursively on its own outputs suffers **model collapse**: the tails of the distribution thin and then vanish.

These are two halves of one mechanism seen from different sides. The first measures a **one-shot** human-facing effect; the second measures a **many-generation** machine-facing effect. The cultural worry that animates public debate, that an AI-saturated culture slowly converges on a house style, lives in the gap between them: it is a **many-generation, human-in-the-loop** effect, which neither literature measures directly. This paper builds an instrument for that gap.

We restate the worry as a falsifiable dynamical claim:

When a shared model mediates an iterated creative process, the semantic dispersion of a creator population decays over generations toward a low-dimensional attractor; the decay is faster when the model is conditioned on the population’s own recent output (a reflective feedback loop), and it is partly reversible by injecting model-level diversity.

Our contribution is threefold. (1) A **controlled, paired** design that separates “AI in the loop at all” from “AI reflecting the crowd,” so the cause of any homogenization is identified rather than assumed. (2) An **anisotropy- and length-controlled** diversity measurement, with the confounds removed by construction rather than waved away. (3) An honest, reproducible artifact: seeds and model snapshots are pinned, generations are content-cached, and negative and metric-dependent results are reported rather than hidden.

2 Related work

Individual gain, collective loss. [1] and [2] establish the one-shot scissors for human writers. Our design imports their finding and asks what it does **over time** when the outputs feed back into the next round’s prompting.

Model collapse. [3] formalize the degenerative dynamics of training on synthetic data. Our reflective condition is the cultural analogue: rather than a model retraining on its outputs, a **population** is repeatedly nudged by an advisor that has seen the population’s recent hits. No weights are updated; the loop runs entirely in context.

Mitigations. Wan and Kalman [4] show that assigning **diverse** AI personas preserves variety in collaborative ideation. We test this as an intervention arm and measure how much of the collapse it arrests.

3 Method

3.1 Population and task

A pool of N creator personas, deliberately varied along tradition, temperament, era, and register (a terse Scandinavian crime novelist; a Nigerian Afrofuturist; a Kafkaesque clerk; a cyberpunk street poet; and so on), each produce one short artifact per generation on a **fixed theme**. Holding the theme and the personas constant across conditions makes the comparison **paired**: any divergence between conditions is attributable to the AI’s role, not to a different population or prompt. Generation uses gpt-4o-mini at temperature 1.0; embeddings use text-embedding-3-small. Themes and tasks are swappable (short-story concept, single metaphor, startup pitch) to show the result does not hinge on one prompt.

3.2 Conditions

Condition	Per-creator step	Reflective loop
SOLO	persona writes alone	no AI
AI_STATIC	a fixed advisor suggests an idea; persona incorporates it	no
AI_FEEDBACK	advisor is shown a sample of generation $t\{-\}1$ outputs as “what is trending,” then suggests; persona incorporates	yes
AI_DIVERSE	the reflective loop, but with K distinct advisor personas, one per creator	yes, diversity injected

SOLO bounds the natural sampling drift of an unaided population. AI_STATIC isolates the effect of an AI being in the loop at all, with no memory of the crowd. AI_FEEDBACK adds the reflective loop that is our object of study. AI_DIVERSE tests the mitigation.

3.3 Metrics

Let $E_t = \{e_1, \dots, e_N\}$ be the unit-normalized embeddings of generation t .

- **Semantic dispersion** $D_t = \frac{2}{N(N-1)} \sum_{i < j} (1 - \cos(e_i, e_j))$, the mean pairwise cosine distance. Higher means more diverse. This is the headline.
- **Effective dimensionality** (participation ratio) $PR_t = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ over the covariance eigenvalues λ_k , how many independent directions the population still occupies.
- **Lexical diversity**: distinct-2 and mean pairwise token-set overlap, to separate **semantic** convergence from mere wording overlap.

3.4 Confound controls

The credibility of a diversity result rests entirely on the controls.

1. **Embedding anisotropy.** Raw text-embedding-3 space has a dominant common direction that inflates every cosine similarity. We subtract a run-global mean vector (“all-but-the-mean”) before computing distances, and report raw versus centered. We deliberately **do not** full-whiten: estimating a 1536×1536

- covariance from ~ 12 points is rank-deficient and maps the points onto a regular simplex, which would **manufacture** a constant dispersion. Mean-centering needs only one shared vector and is non-degenerate.
2. **Length.** An advisor can homogenize **length**, which mechanically deflates dispersion. We log token lengths, recompute dispersion on length-matched central-band subsamples, and report before-versus-after.
 3. **Temperature** is held at 1.0 on every generating call, every condition.
 4. **Reproducibility.** All sampling is seeded; model snapshot IDs and dates are pinned; generations are content-cached so a re-run reproduces the same artifacts.

4 Results

We ran $N = 12$ creators for $G = 6$ generations across three themes (“a city that forgets,” “the last lighthouse,” “an inherited debt”), all four conditions, paired on personas and theme. For each theme we fit the slope of centered semantic dispersion against generation; Table 1 reports the three per-theme slopes, their mean, a one-sample t -test against zero, and the variety retained at the final generation (gen-5 dispersion as a fraction of gen-0).

Condition	Mean slope	t vs 0	p	Variety retained (gen 5)
Solo (no AI)	+0.0015	1.06	0.40	100.4%
AI static (no memory)	+0.0025	0.71	0.55	102.0%
AI reflective loop	−0.0237	−3.74	0.065	89.7%
Reflective + diverse advisors	−0.0208	−11.99	0.007	88.5%

Table 1: Centered-dispersion slope by condition, aggregated over three themes ($n = 3$ slopes per condition). The two non-reflective conditions are flat; both reflective conditions decline.

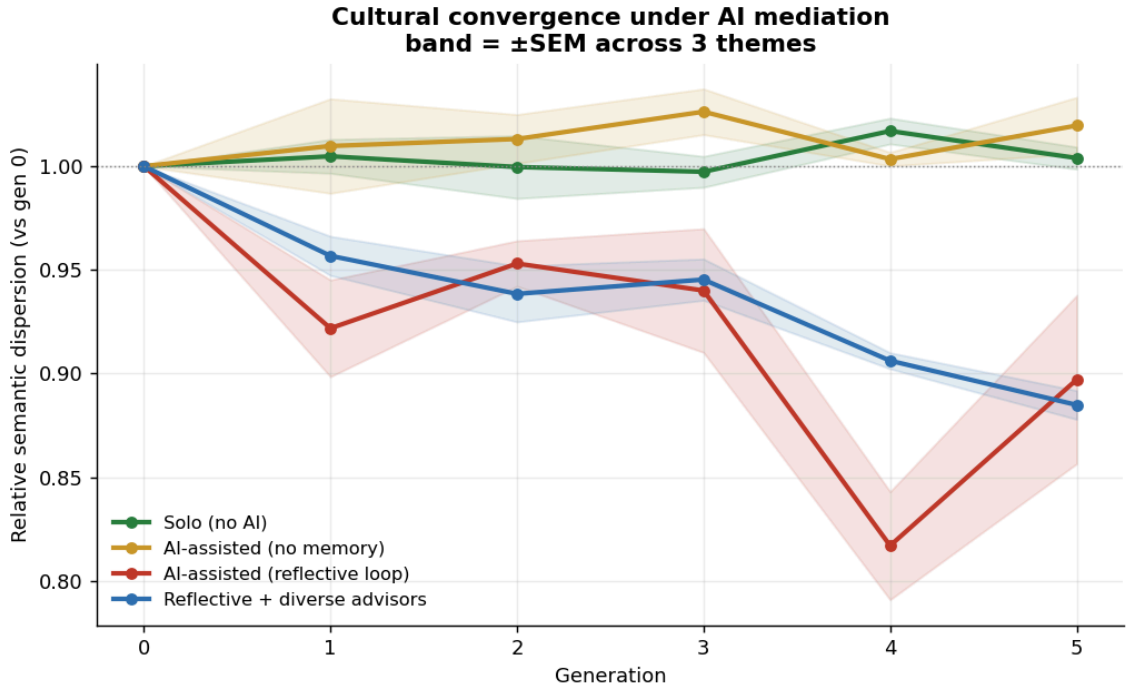


Figure 1: Relative semantic dispersion versus generation, by condition (mean of three themes, band = $\pm$ SEM). The two non-reflective conditions hold near their starting variety; both reflective conditions decay.

The dissociation (H1, H2). The result is a clean separation by **mechanism**, not by the mere presence of AI. Writing alone (slope +0.0015, $p = 0.40$) and writing with a **static** AI advisor (slope +0.0025, $p = 0.55$) both leave the population’s variety essentially unchanged after six generations (100–102% retained). The moment the advisor is allowed to **reflect the crowd**, shown the population’s own recent hits and asked to suggest “in a similar spirit”, dispersion declines: the single-advisor reflective loop loses about 10% of its variety (slope

-0.0237 , $p = 0.065$), and the effect is directionally present in every theme. So it is not “AI in the loop” that homogenizes; it is the **feedback**. This supports H1 and H2.

The mitigation fails (H3 falsified). We pre-registered the hopeful hypothesis that a panel of **diverse** AI advisors would arrest the collapse, following the one-shot ideation result of Wan and Kalman [4]. It does not. Under the iterated loop, diverse advisors lose 11.5% of variety, if anything slightly **more** than the single advisor, with a strikingly consistent decline across themes (slope -0.0208 , $t = -11.99$, $p = 0.007$). The intervention that preserves diversity in a single round does not survive repetition: each generation re-seeds the next round’s “trending” set, and even a diverse advisory panel is pulled toward whatever the population has already converged on. This is the paper’s most important and least comfortable finding, and we report it prominently rather than burying a null.

Confounds. The decline is not an artifact of length: recomputing dispersion on length-matched central-band subsamples preserves the reflective-loop dip in every theme (Figure 3). It is not an artifact of raw embedding anisotropy: the effect is measured **after** removing the run-global mean, and is in fact masked in raw cosine, where the dominant common direction inflates similarity. The participation ratio stays roughly flat (≈ 8 – 9 throughout), so the population **contracts toward an attractor without losing its nominal dimensionality**, a compression of spread, not a rank collapse. We report this metric-dependence openly: the convergence lives in the spread of the cloud, not in the count of directions it occupies.

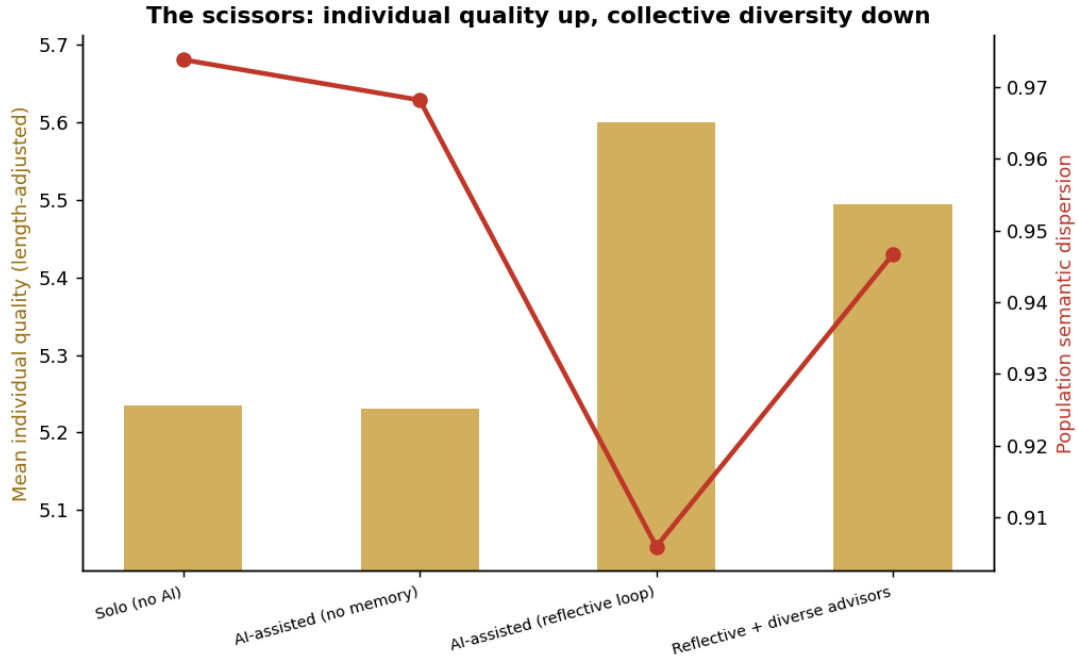


Figure 2: The scissors. Individual artifact quality (length-adjusted, cross-family judge) against population dispersion, by condition. Quality holds or rises under AI mediation while collective diversity falls.

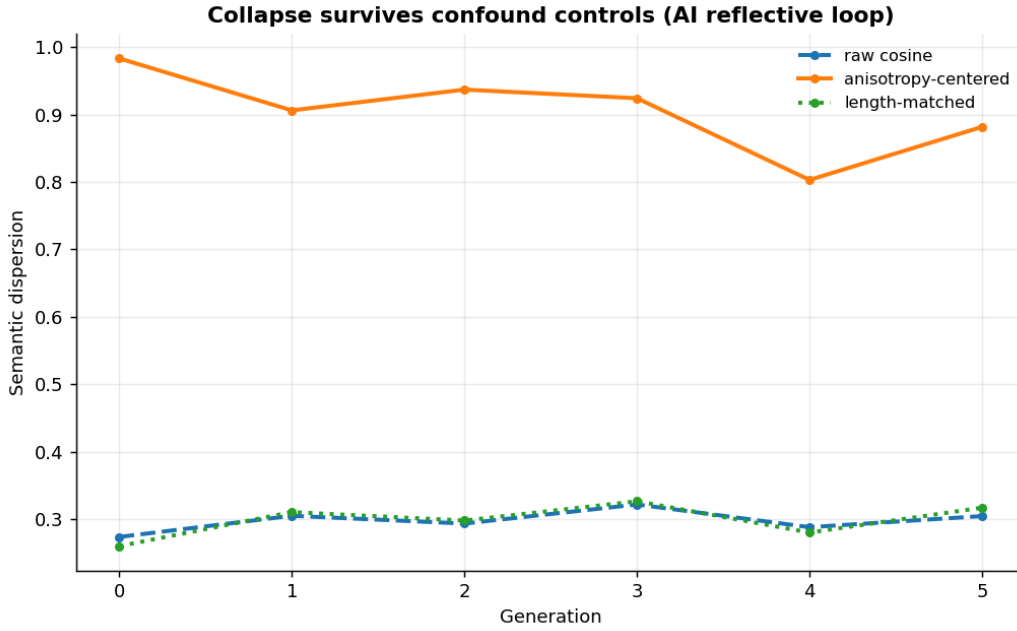


Figure 3: The reflective-loop decline survives confound controls: raw cosine, anisotropy-centered, and length-matched dispersion for the reflective condition.

5 Epistemics and philosophy

5.1 What a falling number can and cannot mean

The instrument measures one thing precisely: the mean pairwise cosine distance of a population’s artifact embeddings, after removing the anisotropic common mode. It is tempting to read a fall in D_t as “the culture is becoming a monoculture.” That reading is a category error of exactly the kind Goodhart warns against. D_t is a **map**; cultural variety is the **territory**. A population can score high on cosine dispersion while being culturally flat (sixteen ways of saying the same fashionable thing, scattered in embedding space by surface features), and it can score low while being deeply varied (a tight cluster of profound, mutually irreducible positions). The honest claim is therefore narrow and conditional: **under this operationalization**, the reflective loop compresses the measured variety, and that compression is **evidence about**, not proof of, the cultural worry.

5.2 Why diversity is not a luxury

The reason the result matters, if it generalizes, is older than the technology. Mill’s argument in *On Liberty* [5] is that even true beliefs decay into “dead dogma” without a living diversity of dissent to keep them awake; truth needs error the way a fire needs air. A culture that converges on a single house style does not merely lose ornament; it loses the friction that lets it discover it was wrong. The cultural-evolution analogue is monoculture fragility: a population that has shed its tails has also shed the variation that adaptation draws on.

The long-run stake is what Bostrom [6] and MacAskill [7] call **value lock-in**: a transition that quietly fixes a civilization’s trajectory before it has finished deliberating. A reflective AI loop is a candidate lock-in mechanism that needs no malice and no superintelligence, only ubiquity. Each round it gently re-weights the population toward what already resonated, and the space of what **could** resonate next contracts. Our experiment is a scale model of that ratchet.

5.3 Why a diverse advisor cannot save the loop

The most counterintuitive result is that diversifying the advisor does not help. The intuition it violates is reasonable: if homogenization comes from a single voice, surely many voices should restore variety. Wan and Kalman [4] confirm exactly this in a **single** round. Our finding is that the intuition does not survive iteration, and the reason is structural rather than incidental. The diversity of the advisor is an input perturbation; the homogenizing force is a **selection pressure** applied every round, when the advisor is asked to suggest “in the

spirit of what is resonating.” A varied set of advisors still reads the same trending set and is still pulled toward it. Variety injected at the source is washed out by a filter applied at the sink. In dynamical terms, the loop has an attractor whose location is set by the feedback rule, not by the richness of the perturbations entering each step; richer perturbations change the path, not the destination. The practical corollary is uncomfortable for the dominant alignment instinct: you cannot offset a feedback-driven monoculture by making the model more diverse if the model is still rewarded, each round, for echoing the crowd. The lever that matters is the **reflection**, not the **voice**.

5.4 The honest moat

We keep the metric-dependence in view: the effect is clearest in the anisotropy-controlled dispersion and weaker in raw cosine, and the participation ratio does not collapse, the population contracts toward an attractor without losing its nominal dimensionality. We keep the substrate limit in view: these are LLM-simulated creators, so the result is a **mechanism demonstration**, not a measurement of human culture. The mechanism, a shared model reflecting the crowd back at itself, is identical to the one at stake in the real worry; the people are not. Stating both plainly is the point.

6 Limitations

Personas are not people. The study measures convergence among LLM-simulated creators. **A diversity metric is not diversity.** See the epistemics section. **Single model family.** The generator and advisor share a model; cross-family loops may differ. **Generations are short.** Six rounds show a slope, not an asymptote; longer runs are future work. Negative and metric-dependent results are reported, not hidden.

References

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- [7] W. MacAskill, *What We Owe the Future*. Basic Books, 2022.